

SYSTEM-H on NATQ-002 validation — the single preregistered run

Decision **FAIL** · 33/33 identity gates · 1 of 3 validation runs consumed · 0 retries · reserve 0 bytes · head `ac25a6c` · 2026-09-05 18:18Z

SYSTEM-H beat the lexical control decisively and still failed the run. Both halves of that sentence are load-bearing. The preregistered paired test came back **+0.1477 micro-MRR** with a 95% interval of **[0.0438, 0.2538]**, which excludes zero — the improvement over BM25 is real and not a resampling artefact. But the decision also requires an absolute floor, and `case_hit@10 = 0.575` is below the **0.65** FAIL threshold, well short of the 0.8 PASS floor. Relative progress, absolute non-qualification. Neither half cancels the other.

FAIL
DECISION

0.575
CASE_HIT@10 · FLOOR 0.65

+0.148
PAIRED MICRO-MRR DELTA

33/33
IDENTITY GATES

1
RUNS CONSUMED

1 — Decision under the replacement rule

PASS clause satisfied: **False** · FAIL clause satisfied: **True** · interval excludes zero: True · mean delta sign: positive.

This FAIL owes nothing to the rule repair. `case_hit@10 = 0.575` is below the 0.65 FAIL floor, a disjunct present in BOTH the superseded and the replacement rule. The repaired regression clause did not affect this outcome and is not what produced the FAIL. The regression disjunct added in EVAL-NATQ2-H-002 changes 5 states of the rule's 36-state space, and this run is in none of them — its delta is positive. The repair was still worth making before the run rather than after, but it is not what produced this outcome, and presenting it as though it were would misstate the record.

2 — SYSTEM-H versus the frozen BM25 comparator

Metric	SYSTEM-H	BM25 control	Delta
micro-MRR (primary)	0.2899	0.1425	+0.1474
case_hit@10 — ANY gold span	0.575	0.375	+0.2
case_full_coverage@10 — EVERY gold span	0.55	0.325	+0.225
span_recall@10	0.5614	0.3333	+0.2281
case_hit@1	0.2	0.1	+0.1
latency p50 (ms)	5493.326	394.016	+5099.31
latency p95 (ms)	8371.576	633.985	+7737.591

Any-evidence and complete-evidence are different numbers and must not be conflated. `case_hit@10 = 0.575` is the fraction of cases where *at least one* gold span was retrieved. `case_full_coverage@10 = 0.55` is the fraction where *every gold span* was. The qualification floors are defined against the former, which is the more forgiving of the two.

3 — The preregistered paired test

Paired bootstrap over **cases**, 10,000 resamples, seed 20260904, exactly as preregistered. Cases are the resampling unit because spans within a case are not independent.

Quantity	Value
Observed micro-MRR delta (H – BM25)	0.147341
Bootstrap mean delta	0.147718
95% percentile interval	[0.043790, 0.253785]
Interval excludes zero	True

Case-level movement: **11** cases SYSTEM-H hits that BM25 misses (A23, A24, A28, B25, C11, C29, D19, D23, D29, E17, E22); **3** the other way (B09, B28, D27); 12 hit by both; **14 missed by both**. That last figure is the ceiling problem — more than a third of the partition is out of reach of either system at depth 10.

4 — Per authoring slice

Slice	n	spans	H hit@10	BM25	H full cov	H span rec	H MRR
A	8	13	0.75	0.375	0.625	0.5385	0.2097
B	8	12	0.375	0.5	0.375	0.4167	0.2292
C	8	11	0.5	0.25	0.5	0.4545	0.3788
D	8	10	0.625	0.375	0.625	0.7	0.3976
E	8	11	0.625	0.375	0.625	0.7273	0.2639

Eight cases per slice, so a single case moves a slice rate by 0.125. These are reported because they were preregistered, not because any one of them supports a conclusion at this sample size.

5 — One scorer, verified before the run

Both systems were scored by the same module. Before SYSTEM-H was run, that scorer was made to reproduce the frozen comparator from its **stored 400-row ranked output**, offline: BM25 retrieval rerun **False**, database queried **False**. It matched on every metric, on the stored 40-case pass/fail vector (0 mismatches) and on every stored per-case span rank (0 mismatches).

Metric recomputed from stored traces	Recomputed	Frozen	Status
case_hit_at_10	0.375	0.375	match
micro_MRR	0.1425	0.1425	match
case_full_coverage_at_10	0.325	0.325	match
span_recall_at_10	0.3333	0.3333	match
case_hit_at_1	0.1	0.1	match

This matters because METRIC-AUDIT-001 found NATQ-001's `strict_recall@10` and NATQ-002's `case_hit@10` NOT_EQUIVALENT. Scoring SYSTEM-H with the NATQ-001 helpers would have measured a stricter metric and made the comparison against 0.375 meaningless. The shared scorer never consults `section_path`.

6 — What this result does not say

- No gain over NATQ-001's historical 50% is claimed; METRIC-AUDIT-001 returned NOT_EQUIVALENT and the two benchmarks share no cases.
- case_hit@10 measures retrieval evidence coverage, not answer accuracy.
- No release promotion follows from this run.

The historical NATQ-001 figure is not a baseline this run improved on: the metrics are NOT_EQUIVALENT and the two benchmarks share no cases, so no comparison between them exists to win or lose.

7 — Run accounting

- 40 cases, 57 gold spans, depth 10, one pass, 0 retries.
- Traces persisted: ranked_output 400 rows, pools 40 rows, ce_scores 40 rows.
- Validation runs: 1 consumed, 2 remaining under the preregistered cap of 3.
- Mean stage latency (ms): a_pool 1300.72, local_bm25 589.52, ce 3581.53, projection 885.58, blend 0.78.
- Reserve opened: False; access log 0 bytes, hash unchanged before and after.

Artifact	SHA256
EVAL-NATQ2-H-002-PREREGISTRATION.json	e17617d5c4cb1481e989484eb77c9bd7f79e95302ae3cf4afc571eb7930fff25
EVAL-NATQ2-H-001-SUPERSESSION.json	fb589e81e1d62ad9b0a9f9c70136598c0b82a2e791cd595dd1a292206e2bb621
EVAL-NATQ2-H-002-RESULTS.json	ce01aeee9bebd811b9a3ae1a2f58983ca607e3bb8f18096b232a0a2ddafff9ce
EVAL-NATQ2-H-002-CASE-RESULTS.json	cc289dcbe10807330df8527d3f1313ce08a3c5e41e196e89a9239408ad0371ff
EVAL-NATQ2-H-002-RUN-ACCOUNTING.json	4c22098985f437b4ca8b7b13d9483f4bb4616817697f9ce569fe6ead7396f6d2
natq2_scorer.py	96447990b75f94e2f1e3daad1407eb1cdac53f7e4967ceb4ecdc4c0f8119b060
h-ranked-output.jsonl	545eae62ff55bdd5f70ab9f24136546ff5a239c82392e179db0b896ed5bb63a

Authoritative SYSTEM-H config hash `7599eb3c3bb4798230a722e6a7c96b046dc36cc0332b584ec55339896b2d717a` , preserved and gate-checked. SYSTEM-H was not redesigned — the wrapper imports the recovered pipeline verbatim and changes only the input partition and the scorer. No BM25 retrieval rerun, no tuning, no reserve or holdout access, no release promotion. Prior status at head `195b073` .